

Multiple Choice (10 marks)

1. Find the solution for the equation $\dot{X} = AX + bu$; $X(0) = X_0$ where $A = \begin{bmatrix} 10 & 1 \\ -1 & -2 \end{bmatrix}$, $b = 1$, $X_0 = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$, $u(t) = \delta(t)$ (impulse function).
 - (a) -3
 - (b) 2
 - (c) 0
 - (d) 3
 - (e) 10
2. Use partial fraction expansion to find the inverse z-transform of $F(z) = \frac{\{2z^2 - 2z\}}{\{(z - 3)(z - 5)^2\}}$.
 - (a) $f[n] = 4nu[n] + [(5/6)n - 1] 6^n u[n]$
 - (b) $f[n] = 6nu[n] + [(2/3)n - 1] 3^n u[n]$
 - (c) $f[n] = 3^{n+1}u[n] - 5^n u[n]$
 - (d) $f[n] = 3nu[n] + [(4/5)n - 1] 5^n u[n]$
 - (e) $f[n] = [(2/3)n + 1] (3/2)^n u[n] - 2^{n-1} u[n]$
3. A voltage source $60\cos 1000t$ V is in series with a 2-k Ω resistor and a 1 μ F capacitor. Find i_{forced} .
 - (a) $i_{\text{forced}} = 28\cos 1000t - 14 \sin 1000t$ mA
 - (b) $i_{\text{forced}} = 24\cos 1000t - 12 \sin 1000t$ mA
 - (c) $i_{\text{forced}} = 20\cos 1000t + 10 \sin 1000t$ mA
 - (d) $i_{\text{forced}} = 30\cos 1000t - 15 \sin 1000t$ mA
 - (e) $i_{\text{forced}} = 15\cos 1000t - 30 \sin 1000t$ mA
4. The rating of a certain machine from the name-plate is 110 volts, 38.5 amperes, 5 h.p. Find the input, output and efficiency at full load.
 - (a) Input: 4235 watts, Output: 3600 watts, Efficiency: 85 percent
 - (b) Input: 4500 watts, Output: 4000 watts, Efficiency: 88.9 percent
 - (c) Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent
 - (d) Input: 4100 watts, Output: 3450 watts, Efficiency: 84.1 percent
 - (e) Input: 4200 watts, Output: 3900 watts, Efficiency: 86 percent
5. Find the Laplace transform of $f(t) = t^2$.
 - (a) $(2 / s^4)$

- (b) $(1 / s^3)$
- (c) (t^2 / s)
- (d) (t^2 / s^3)
- (e) $(2 / s^3)$

True / False (10 marks)

Answer True or False

6. True or False: In a lap winding DC machine with 100 conductors and 10 parallel paths, the average pitch is 10.
7. True or False: A D-flip-flop is transparent when the output is independent of the input signal.
8. True or False: The function $f(x, y)$ has a removable discontinuity at the origin $(0, 0)$.
9. True or False: The power delivered by a roller chain of $3/4$ in. pitch on a 34-tooth sprocket at 500 rpm is 6.73 horsepower.
10. True or False: XOR and XNOR gates are universal gates that can implement any digital circuit by themselves.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of converting temperature from Celsius to both the Fahrenheit and Kelvin scales, using -10°C as a reference point. Include detailed calculations and implications in engineering contexts.
12. Derive the general solution of the differential equation $d^2x/dt^2 - 2(dx/dt) - 3x = 10\sin(t) + b(2t + 1)$, discussing each step and the roles of constants a and b .

End of Paper